Building and Scaling Inference Workloads on Kubernetes (EKS)



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Welcome



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Patis Piriyahaphan
Solutions Architect,
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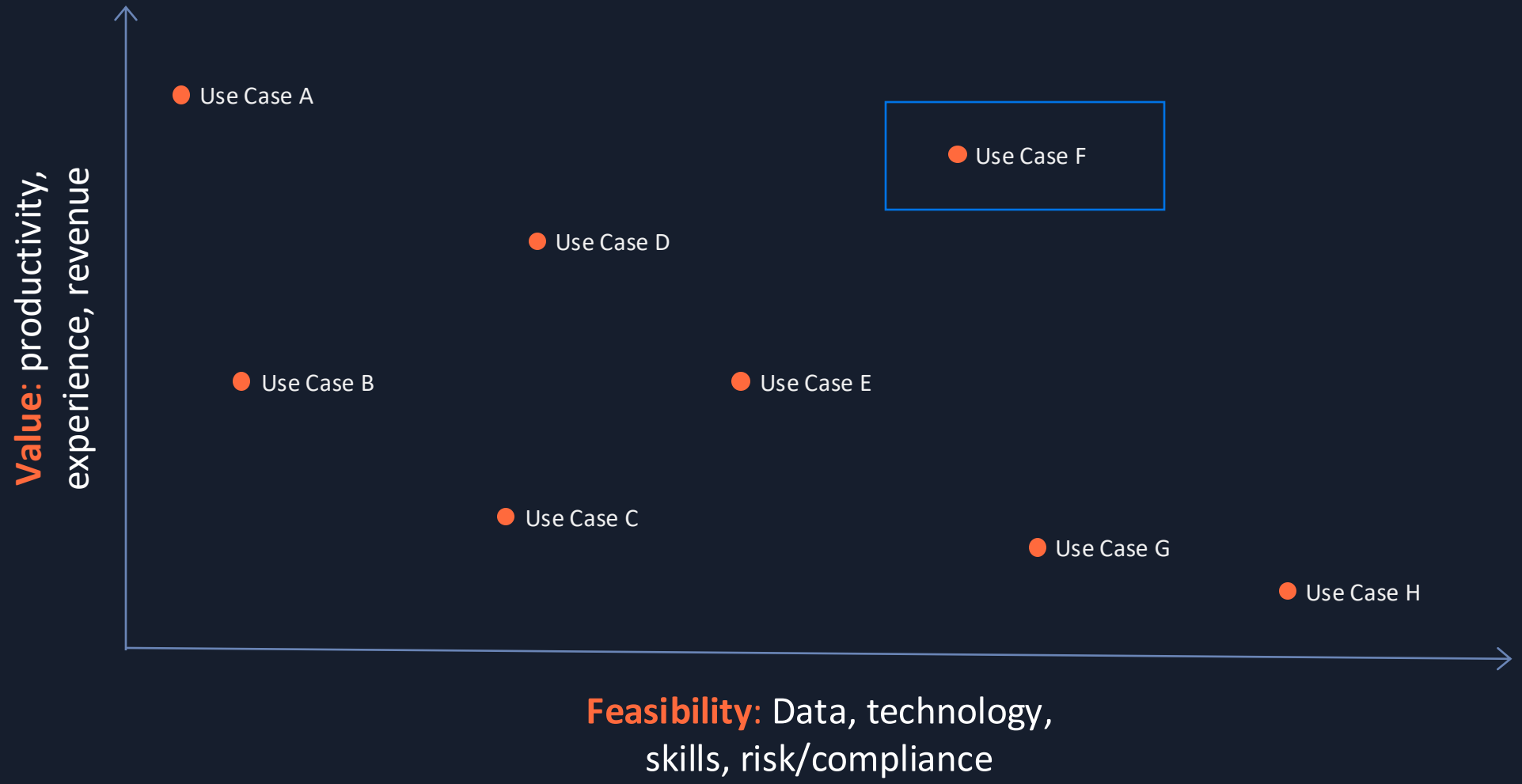
Kajonsak Temeeyajit
Senior Solutions Architect,
Amazon Web Services

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Selecting the right use cases



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Existing Pattern

Build the Model



Training

Use the Model



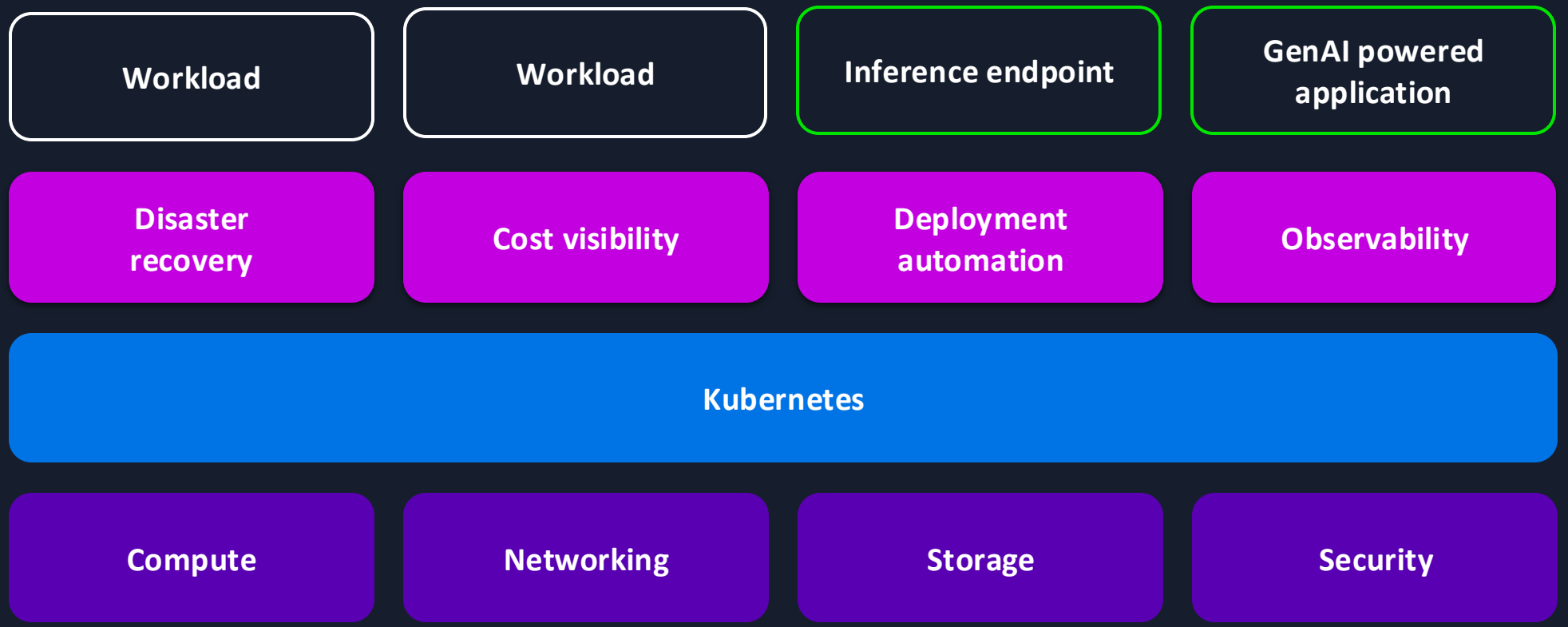
Inference

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So we’re going to self-host the inference

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The best place to self-host is on Kubernetes



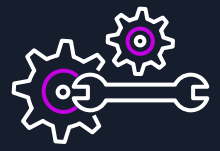
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Why we choose Kubernetes (EKS) for inference



Tooling ecosystem

Vibrant open-source system of tooling



Extensibility

Built-in Kubernetes native integrations with AWS ML infrastructure services



Cost optimization

Control of environment to deploy cost optimized generative AI applications



Configurability

Full control of your data and infrastructure configuration



Highly scalable

Full control of your data and infrastructure configuration



Run anywhere

Consistent deployments across clouds, on-premise, and at the edge

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At the heart of GenAI, is a model

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Anatomy of a Model

Working Model

Model Weights (.bin, .safetensors, .Pt)	The learned parameters used to compute the output through a mathematical operation sequence ~ many GB
Model configuration file (config.json)	Describes the architecture of the model. Used by the Inference engine to load the correct model class
Tokenizer (LLMs) (tokenizer.json)	A process of mapping text to tokens before sending it to the model

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Choose Foundational Model(s)

- With the rise of agentic models, **choice matters** when it comes to model selection.
- Models with performant function calling capabilities – (tau bench benchmarks)
- Cost Evaluation - Certain tasks require more intelligence per $ than others
- Certain tasks require different foundational models – image, video, speech

AI21labs	amazon	ANTHROPIC	cohere	deepseek	Luma	Meta	MISTRAL AI_	poolside	stability.ai
Effective reasoning & rapid analysis for long context windows	Frontier intelligence & industry leading price performance, high-quality image & video generation	Advanced reasoning & coding capabilities, including computer use skills	Multimodal search & advanced retrieval powering multilingual knowledge agents	Advanced reasoning, comprehension, scientific & mathematical analysis, & code generation	High-quality video generation from text & images	Advanced image & language reasoning	Knowledge summarization, expert agents, & code completion	Software engineering AI for large enterprises	High-quality AI image generation, easily deployable at scale
JAMBA	AMAZON NOVA	CLAUDE	COMMAND EMBED RERANK	DEEPSEEK-R1	LUMA RAY 2	LLAMA	MISTRAL MIXTRAL	MALIBU POINT	STABLE DIFFUSION STABLE IMAGE

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Choosing a size

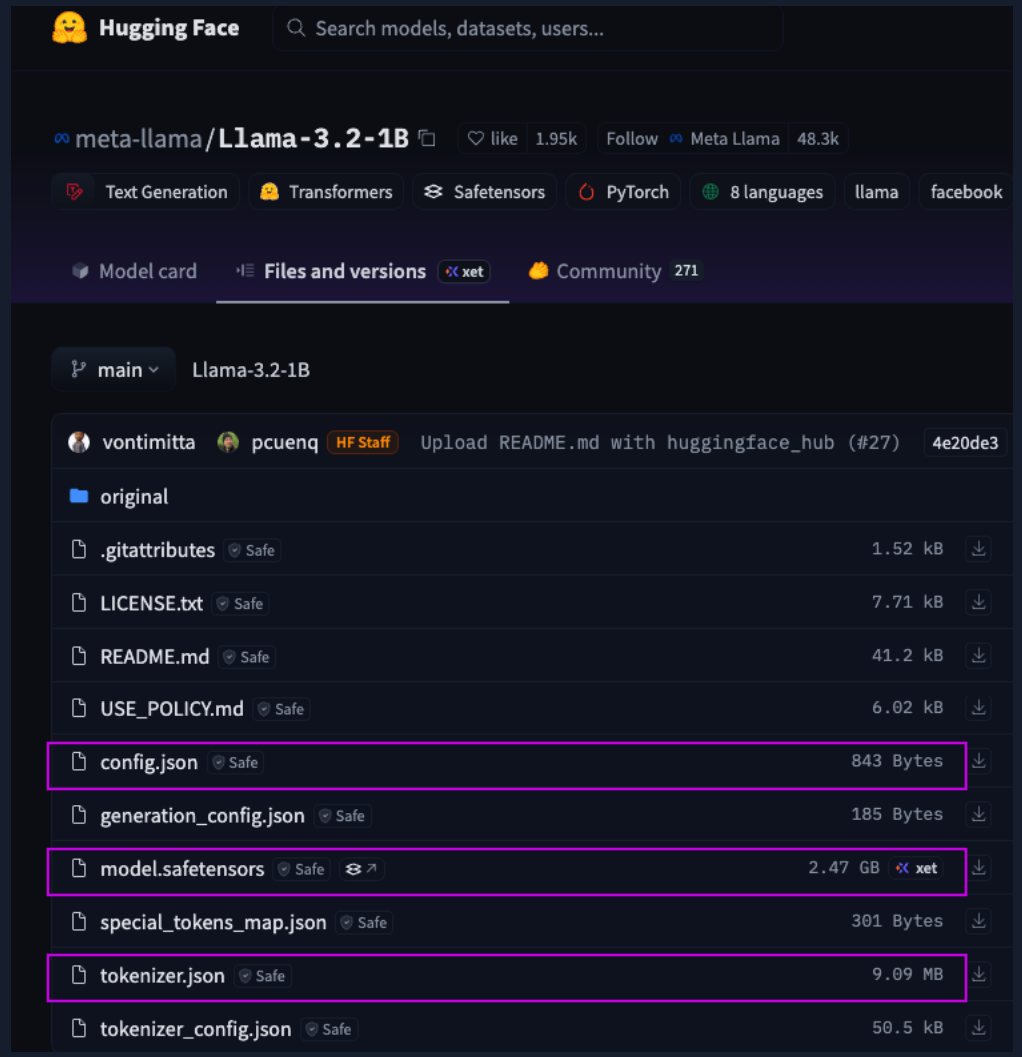
Llama-3.2-1B

or

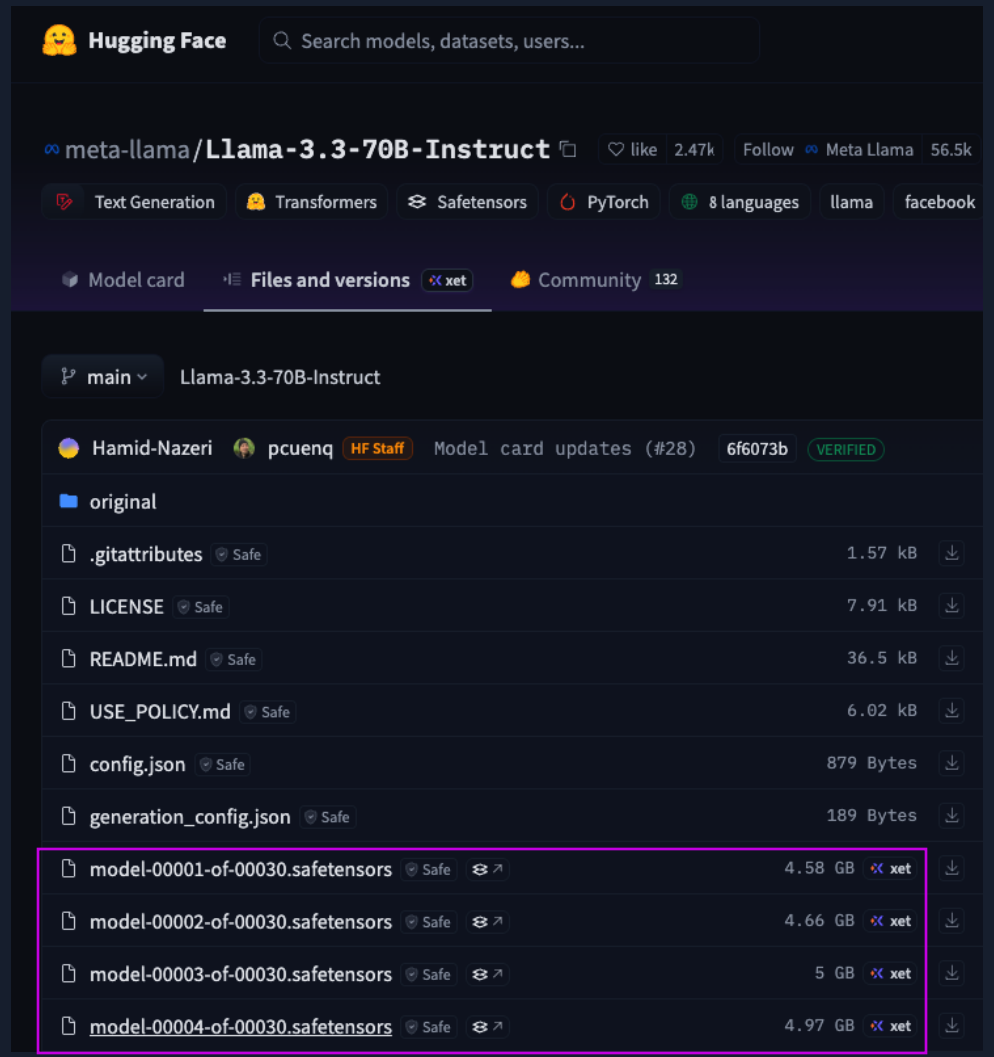
Llama-3.2-70B

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1B vs 70B model size



VS



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Model size determines GPU requirements

Model size = Parameters (70-billion) x 2 bytes (FP16) = 140 GB

Llama 2 – 70B

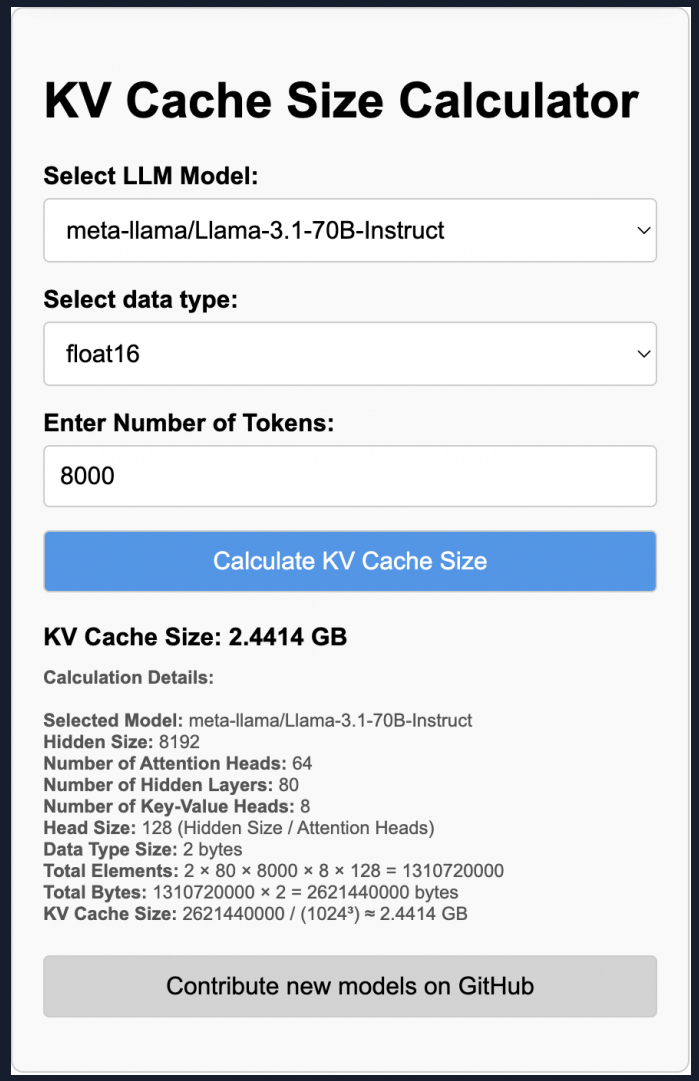


* The required memory depends on input/output sequence length, decoding strategy, model architecture, & batch size

Quantizations:
FP8 = ½ FP 16= 77 GB
FP4 = ¼ FP 16= 39 GB
INT4 …
INT8 …

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KV Cache size calculator



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Accelerated computing portfolio

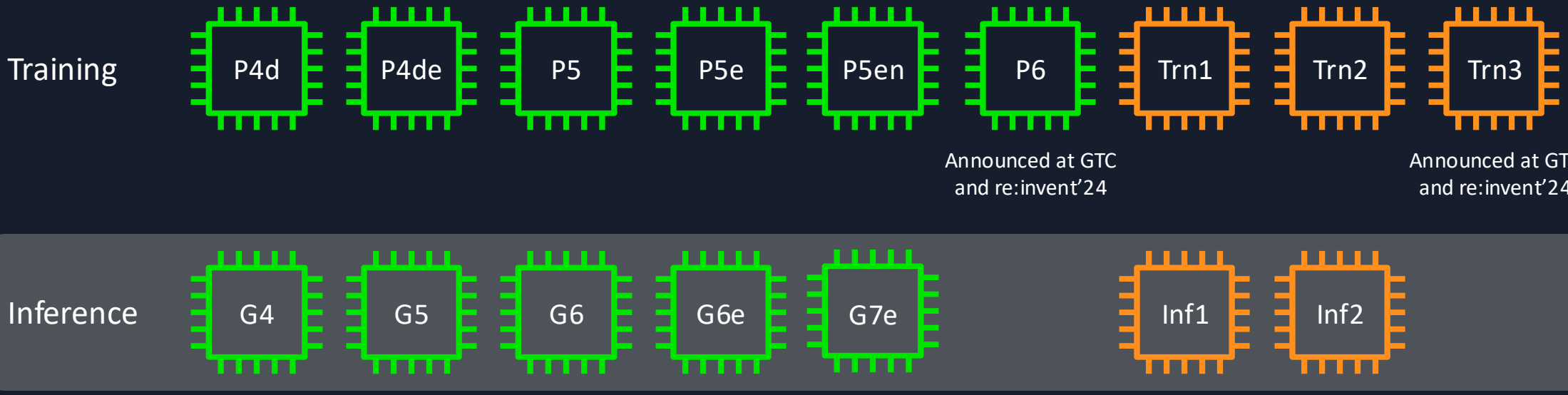
GPU and
AWS ML Accelerators

NVIDIA

H200, H100, A100,
RTX PRO 6000, L4,
L40S, A10G, T4

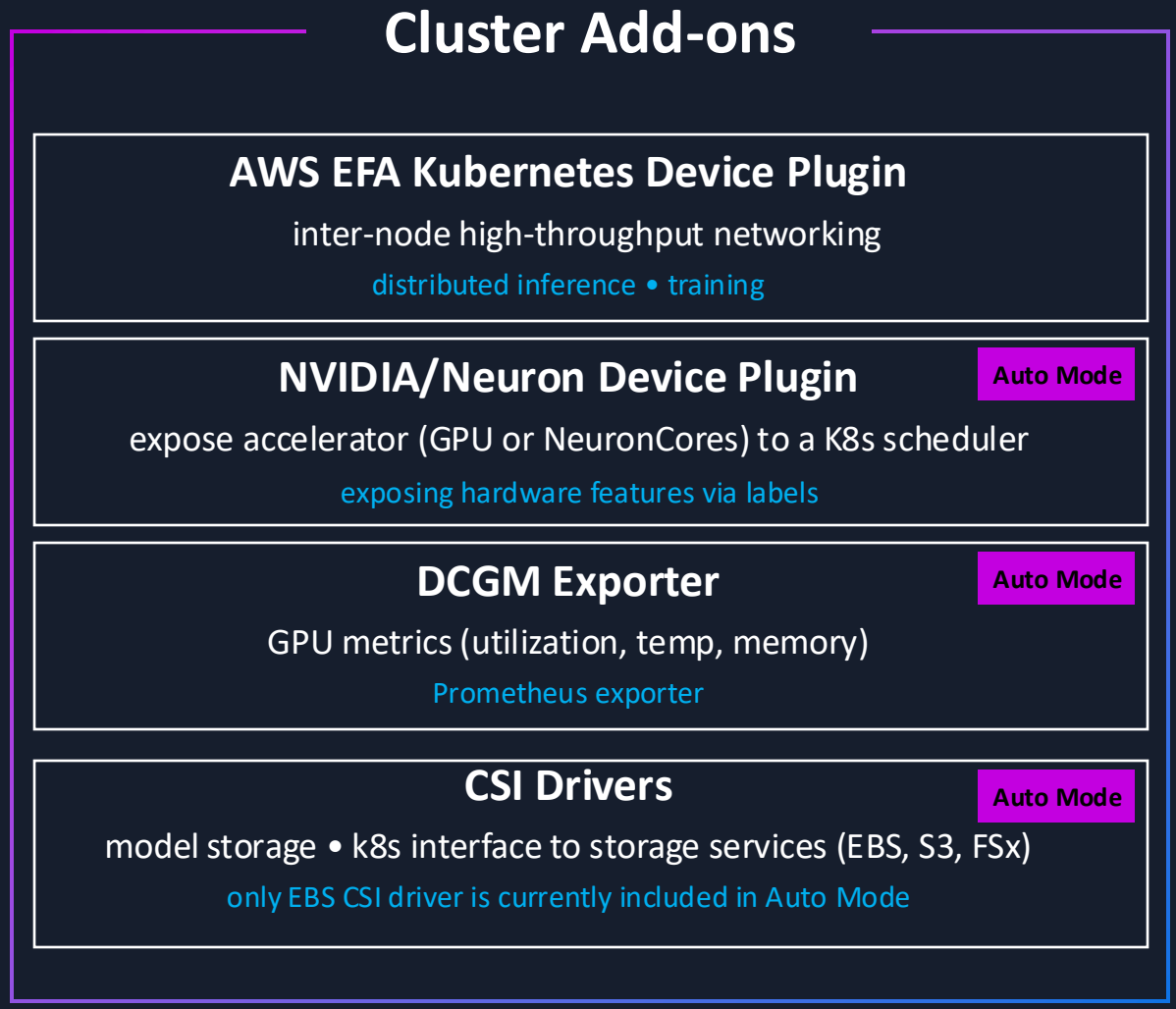
aws

Trainium accelerator
Inferentia accelerator



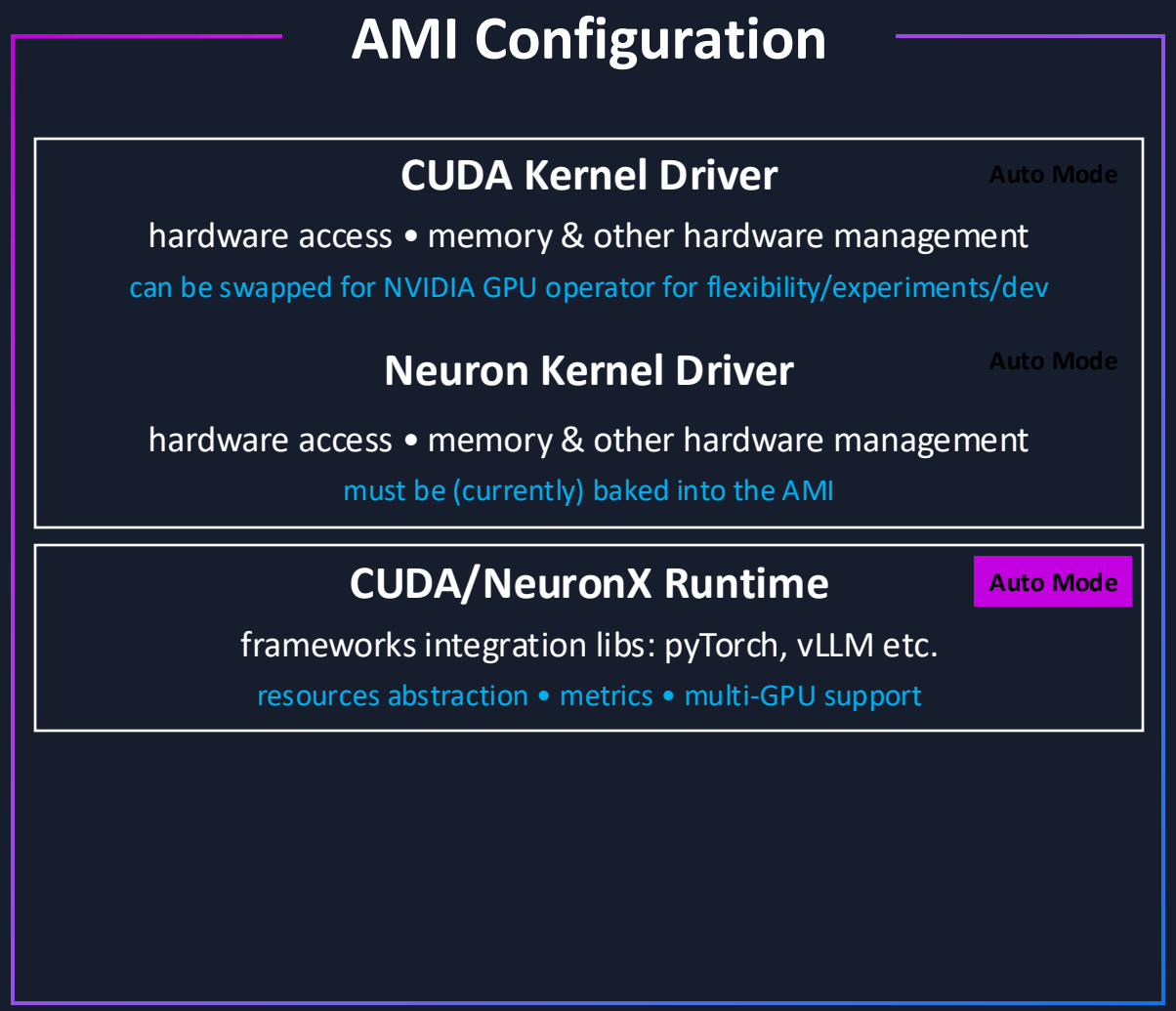
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GenAI specific components: Add-Ons



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GenAI specific components: AMI



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EKS-accelerated AMIs

→ **Device Plugin** can be easily installed as an EKS add-on

Bottlerocket



Device Plugin

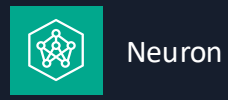
NVIDIA Runtime

NVIDIA kernel driver
with MIG Support

EFA Adapter

Optimized for EKS OS

Bottlerocket



Neuron kernel driver
dkms

EFA Adapter

Optimized for EKS OS

AL2023



NVIDIA Runtime

NVIDIA kernel driver
with MIG Support

EFA Adapter

Optimized for EKS OS

AL2023



Neuron Runtime

Neuron kernel driver
dkms

EFA Adapter

Optimized for EKS OS

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Deep Learning Containers

Multiple ML frameworks support

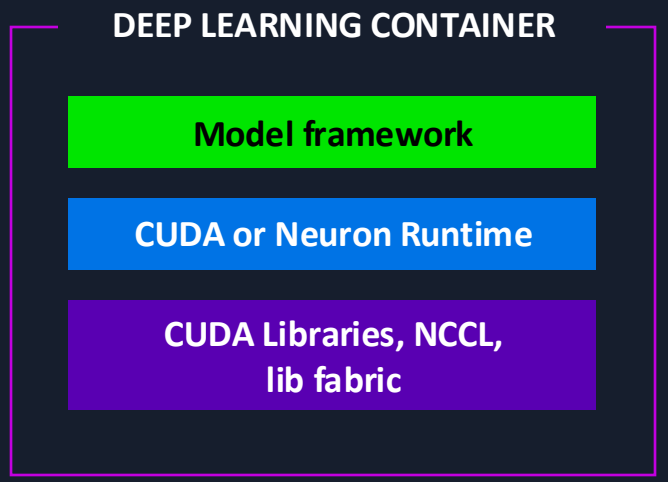
Configured with multiple ML frameworks such as Tensorflow, PyTorch, XGBoost, and JAX. Includes python packages, i.e., numpy, sklearn, scipy, pandas, and many others.

Pre-bundled ML libraries

Includes text generation, text embeddings toolkit, transformers, and diffusers library. Contains Nvidia,
Neuron packages, CUDA, CuDNN, NCCL, vLLM libraries.

OSS integrations

Optimize LLM serving using vLLM. Scale model serving using framework agnostic library such as Ray.



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Inference Engines



- vLLM
- Llma.ccp
- sglang
- TensorRT-LLM
- HuggingFace TGI
- ...

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Inference engines: Compatibility matrix

	vLLM	TensorRT-LLM	TGI
Model Types	**LLMs** **Decoder** LLaMA • Mistral • MixtralFalcon • Qwen	**LLMs** **Decoder** **Encoder** **Multi-modal** LLaMA • Mistral • Mixtral • Qwen • TS • BART • LLaVA • Qwen-VL	**LLMs** **Decoder** LLaMA • Mistral • Falcon • StarCoder
Model Formats	**HuggingFace** **.pt** **.safetensors** direct loading • no conversion needed	**HuggingFace** **.safetensors** export **.onnx** Optimum **.plan** requires compilation • but enables hardware optimization	**HuggingFace** **.safetensors** **.plan** + plugin native HuggingFace ecosystem integration
Multi-GPU Support	**single node** **Tensor Parallelism** weights split across GPU **horizontally**	**multi-node** **Tensor Parallelism** **Pipeline Parallelism** TP: split operations • PP: split layers • ultra-large models	**single node** **Tensor Parallelism** weights split across GPU **horizontally**
Hardware Support	**NVIDIA** **AMD** **Inferentia** **Intel** cross-platform flexibility	**NVIDIA** maximum GPU performance optimization	**NVIDIA** **AMD** **Inferentia** **Intel** cross-platform flexibility

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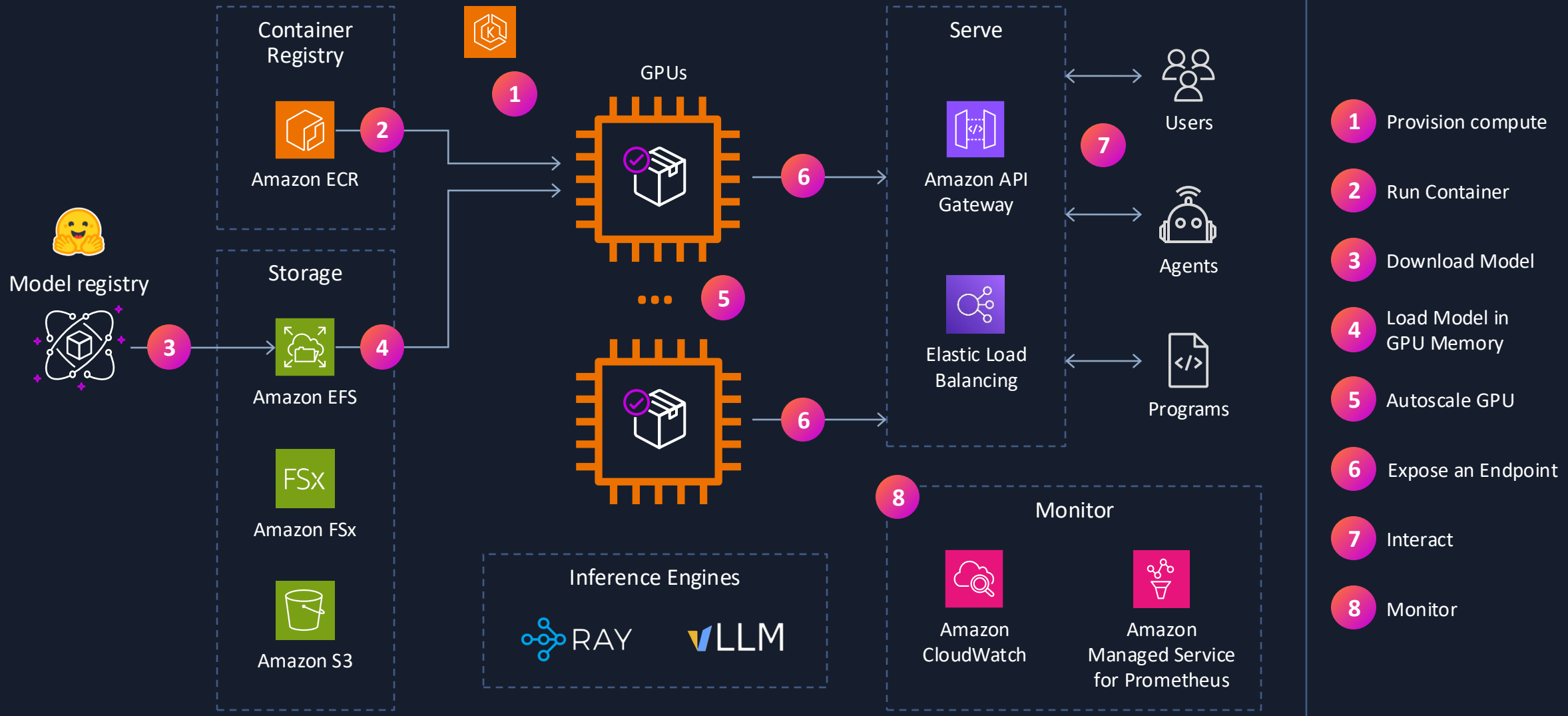
Inference at scale?



- **Continuous batching:** Serves multiple requests simultaneously without waiting for the longest sequence to finish
- **PagedAttention:** Manages KV cache in non-contiguous memory blocks, eliminating fragmentation and maximizing GPU memory utilization
- **Optimized KV cache:** Shares cache across requests with common prefixes; supports FP8 KV for reduced memory footprint
- **High throughput:** Maximizes GPU saturation through dynamic batching and chunked prefill
- **Low latency at high concurrency:** Scales to hundreds of concurrent sequences with minimal per-request overhead

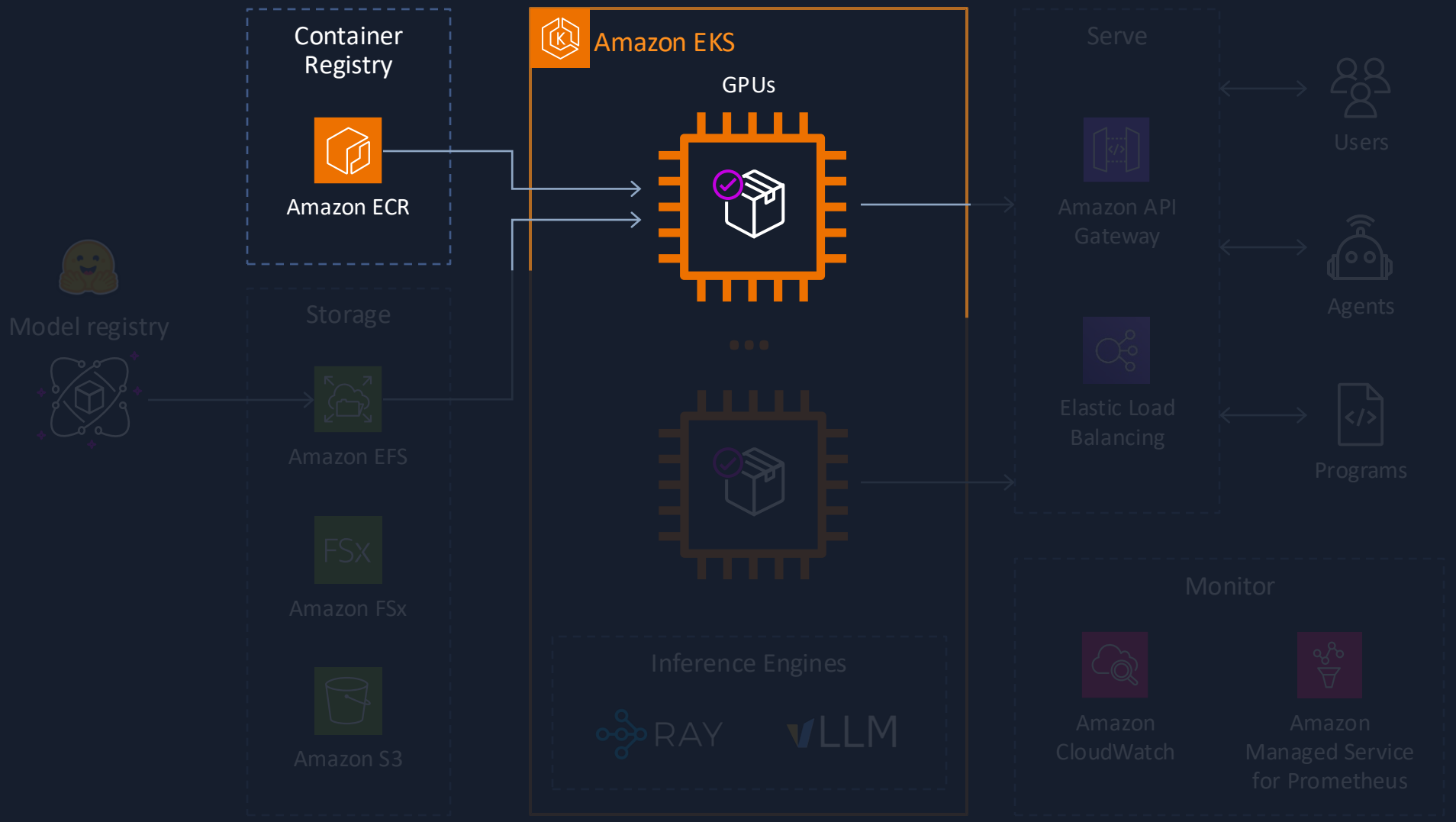
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Inference architecture with auto-scaling



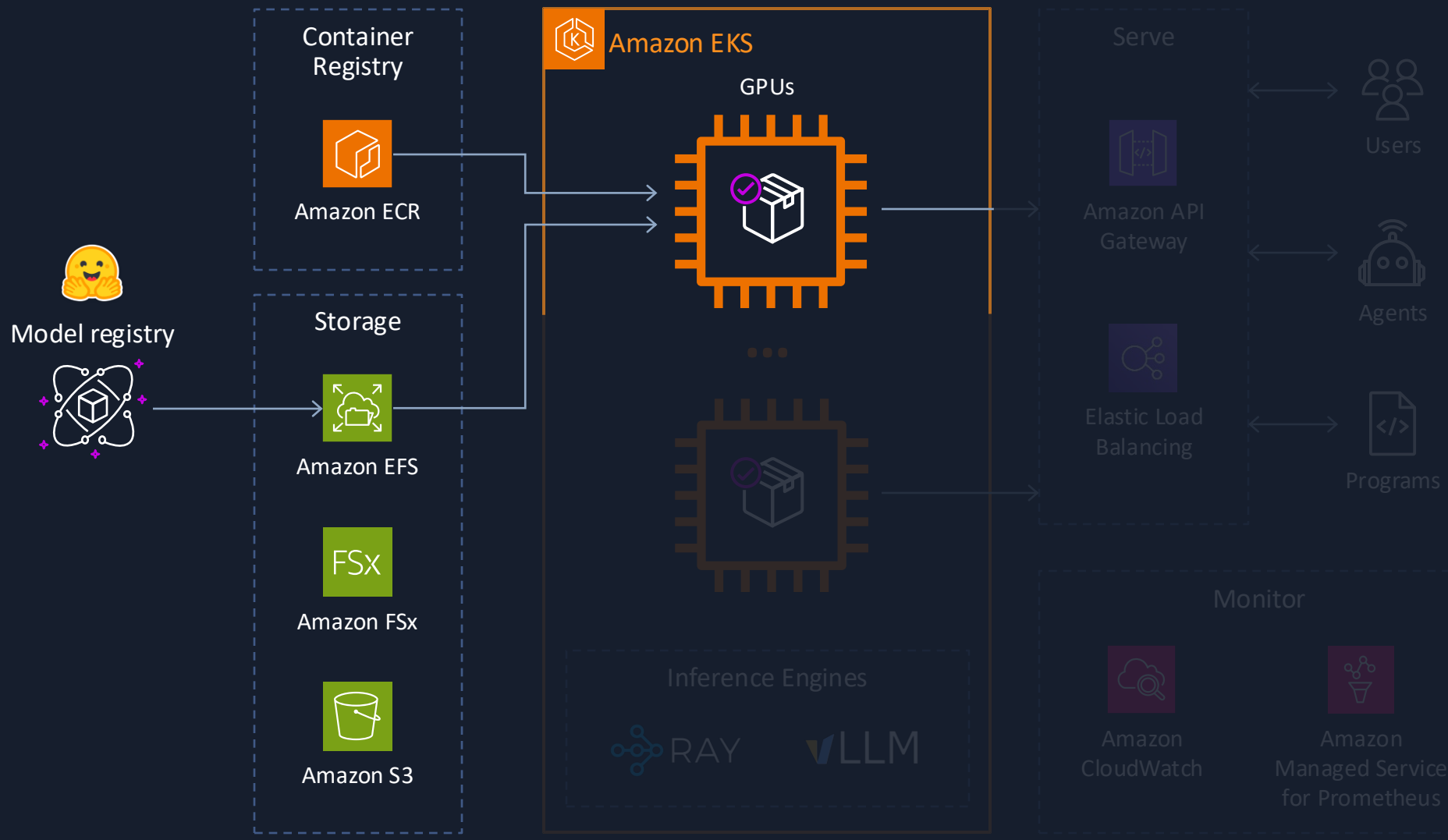
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Deep Learning Containers



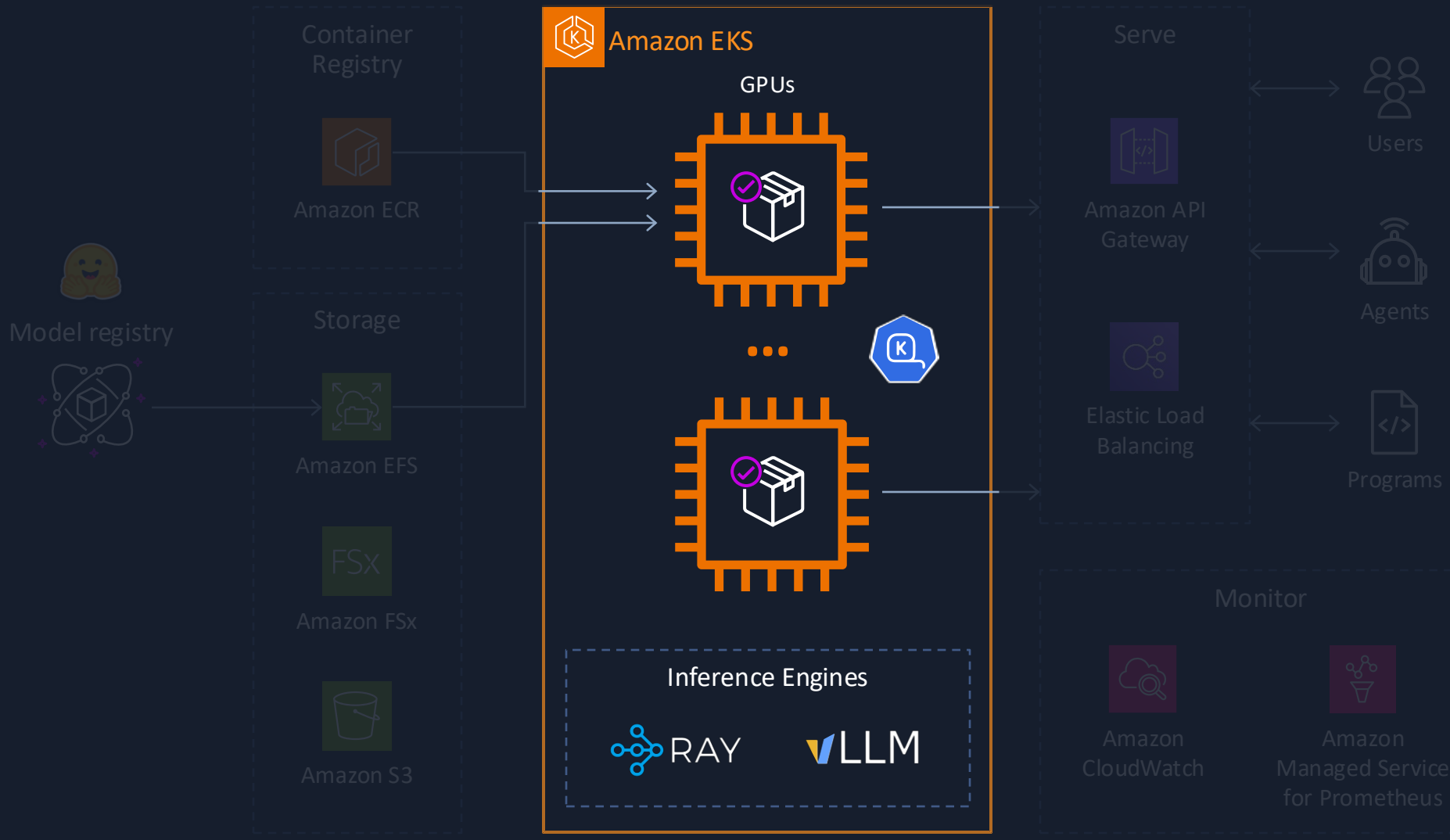
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Load the model into the GPU



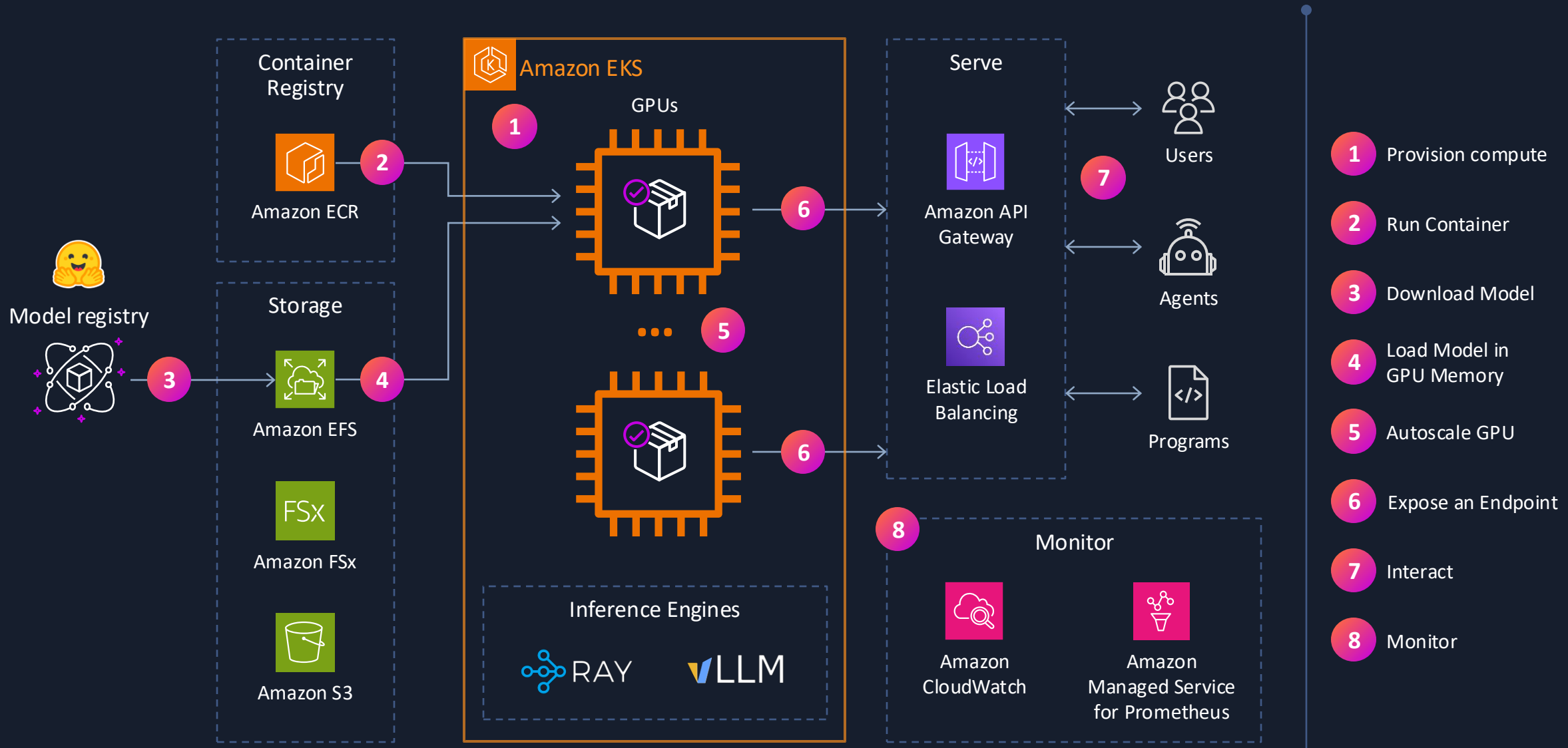
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Auto-scaling GPUs



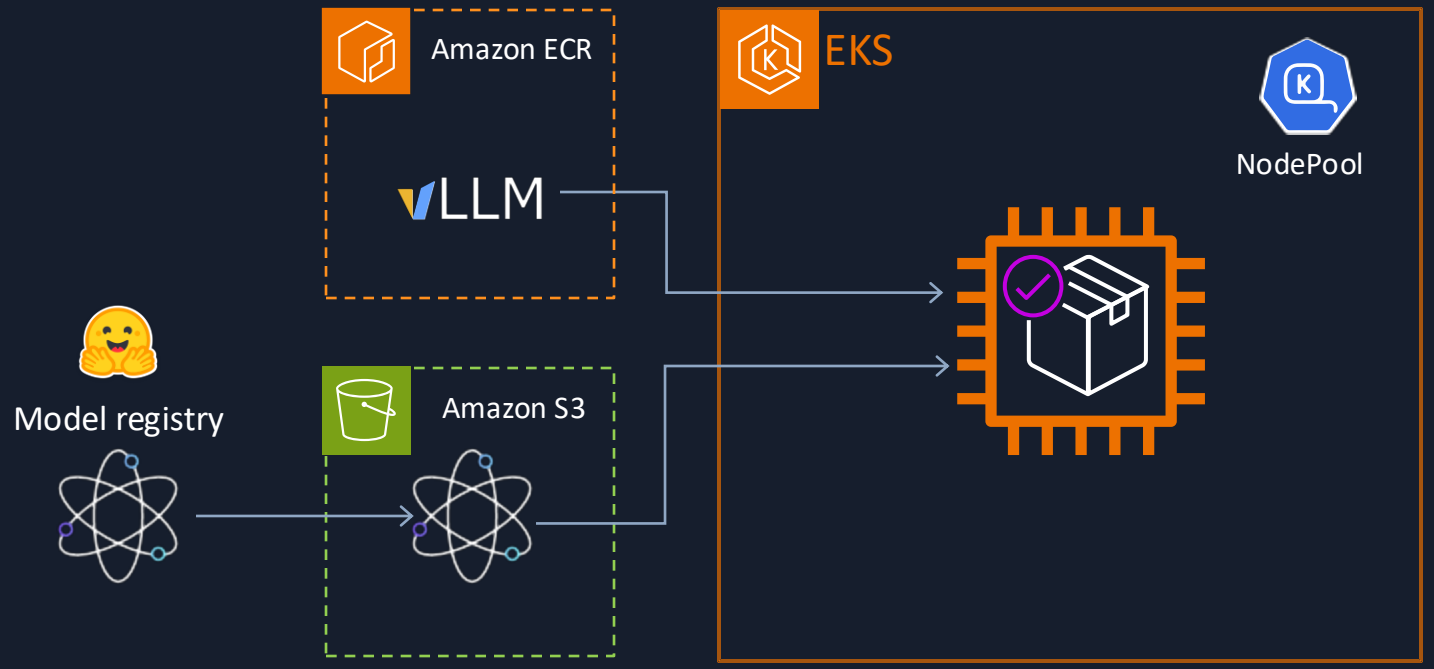
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Inference architecture with auto-scaling



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Run workload

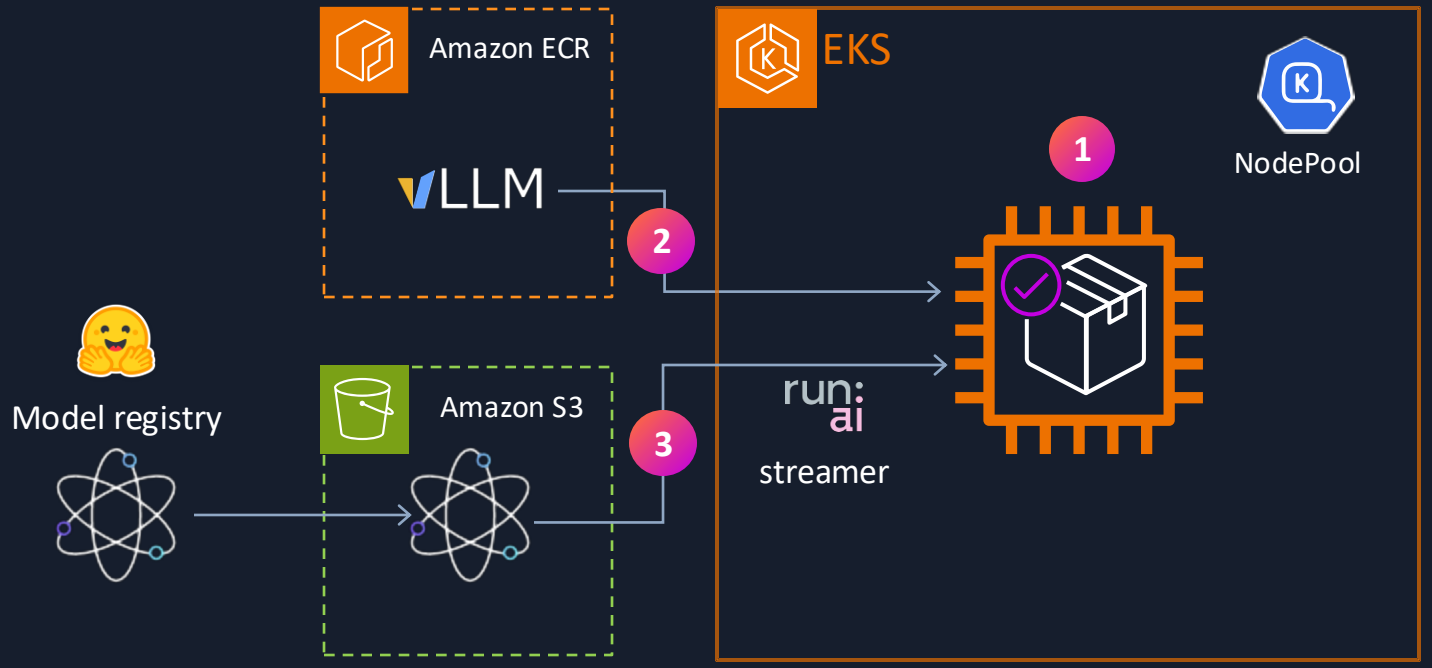



```yaml
apiVersion: apps/v1
kind: Deployment
...
spec:
  template:
    spec:
      containers:
      - name: server
        image: public.ecr.aws/deep-
               learning-containers/vllm
        args:
          - --model /s3/ministral3-8b
          - port "8080"
        env:
        - name: VLLM_ATTENTION_BACKEND
          value: "TRITON_ATTN_VLLM_V1"
...
        nodeSelector:
          intent: gpu
        tolerations:
        - key: nvidia.com/gpu
          operator: Equal
          value: "true"
          effect: NoSchedule
```

```
> kubectl apply -f deployment.yaml
```

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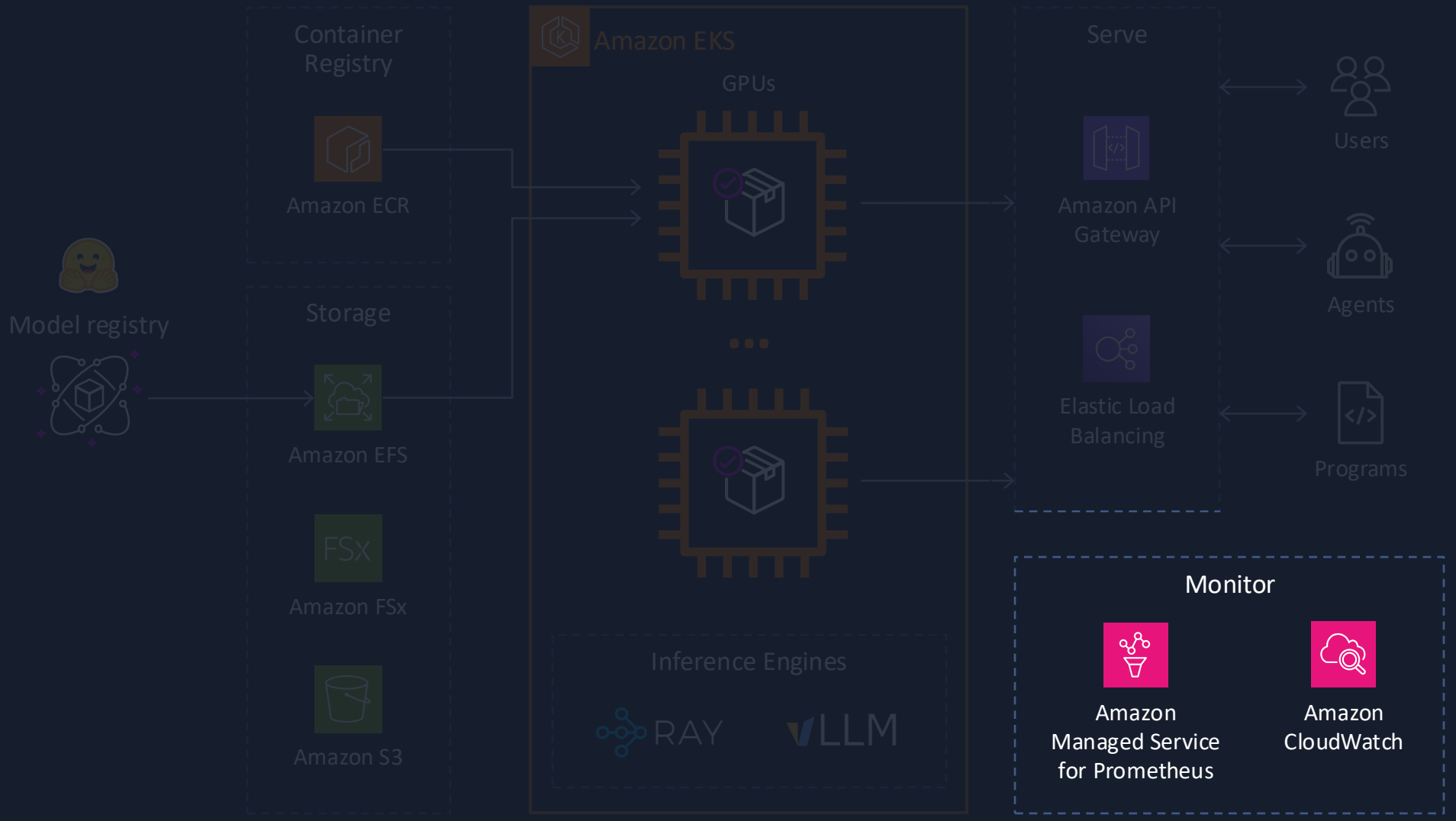
Architecture overview





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Monitoring Your Workloads



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Flexible GPU Observability with Amazon Managed Prometheus

Prometheus

- **Real-Time PromQL Queries:** Query live DCGM metrics for GPU troubleshooting without data lag
- **Long-Term Metric Retention:** Historical GPU telemetry for capacity planning and trend analysis
- **Agentless Scraper:** Managed scraper eliminates node-level resource contention on GPU instances producing high cardinality metrics.
- **Anomaly Detection:** Easily analyze metrics with random cut forest algorithm to detect anomalous behavior in GPU telemetry.

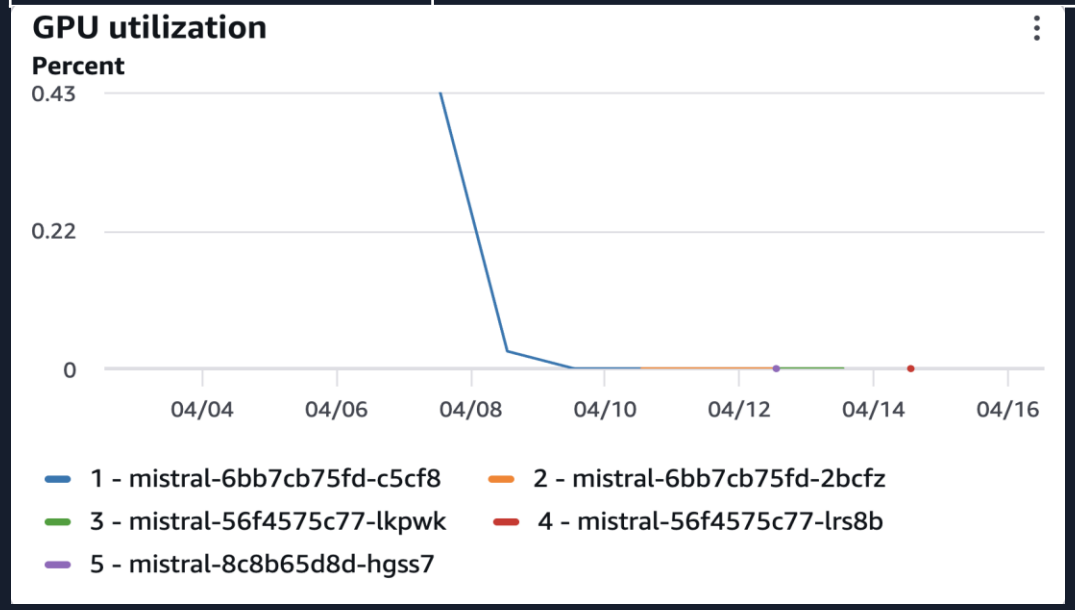
Grafana

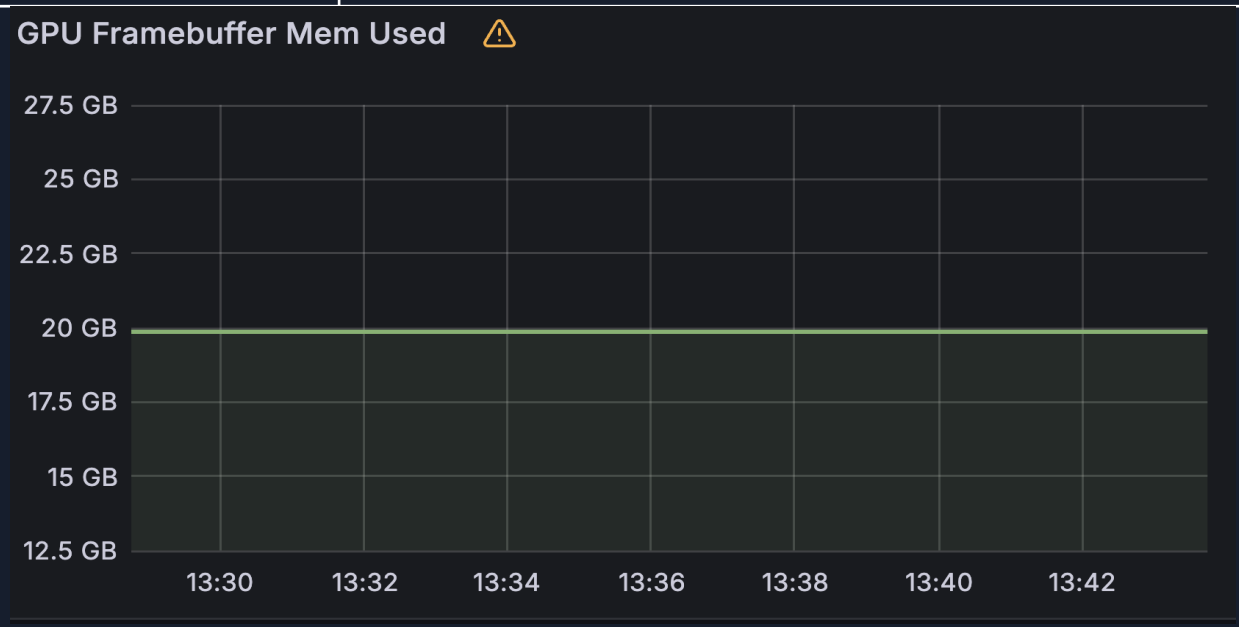
- **Pre-Built DCGM Dashboards:** Out-of-the-box Grafana dashboards for GPU telemetry without custom configuration
- **Custom GPU Alerting:** Threshold-based alerts on GPU anomalies with Amazon Managed Prometheus rules
- **Multi-Cluster Visualization:** Unified Grafana workspace aggregates GPU metrics across EKS clusters
- **Automated Version Updates:** Managed Grafana updates without dashboard migration or downtime

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GPU Metrics to Monitor During Inference

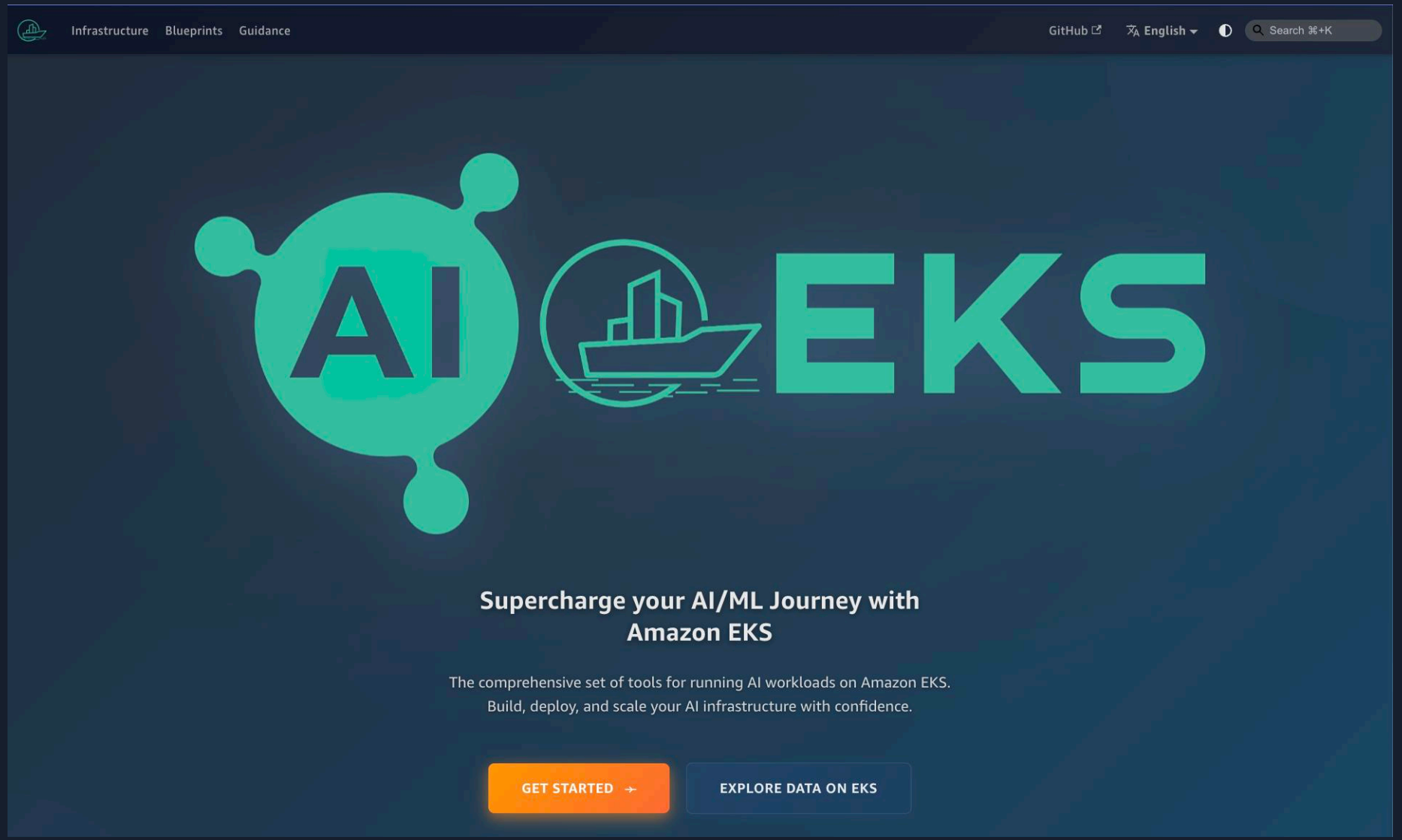
Metric	What It Tells You	Why It's Important
GPU Utilization	Percentage of time GPU is actively processing	Shows if GPUs are efficiently used or over/under-provisioned
GPU Power Consumption	How much power the GPU is drawing	Low power at high utilization signals inefficient workload usage
GPU Framebuffer Mem Used	Actual memory consumption in megabytes	Tracks resource usage for capacity planning
GPU Memory Total	Total framebuffer size available	Defines capacity ceiling for sizing decisions





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How to get started



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Demo

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EC2 G7e instance
NVIDIA RTX PRO 6000
ับ MiniMax

Key takeaways



Flexibility

Run open or proprietary models on EKS using your preferred frameworks, with **no vendor lock-in** and **easy experimentation**

Scalability

Scale inference workloads **dynamically** with EKS and OSS tools like Karpenter and Ray **based on traffic**

Performance

Direct GPU access on EKS, combined with efficient utilization and autoscaling, delivers **low-latency** inference at **low cost**

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To learn more

Inference on EKS Workshop

AI/ML on EKS user guide

AI/ML on EKS Terraform Blueprints

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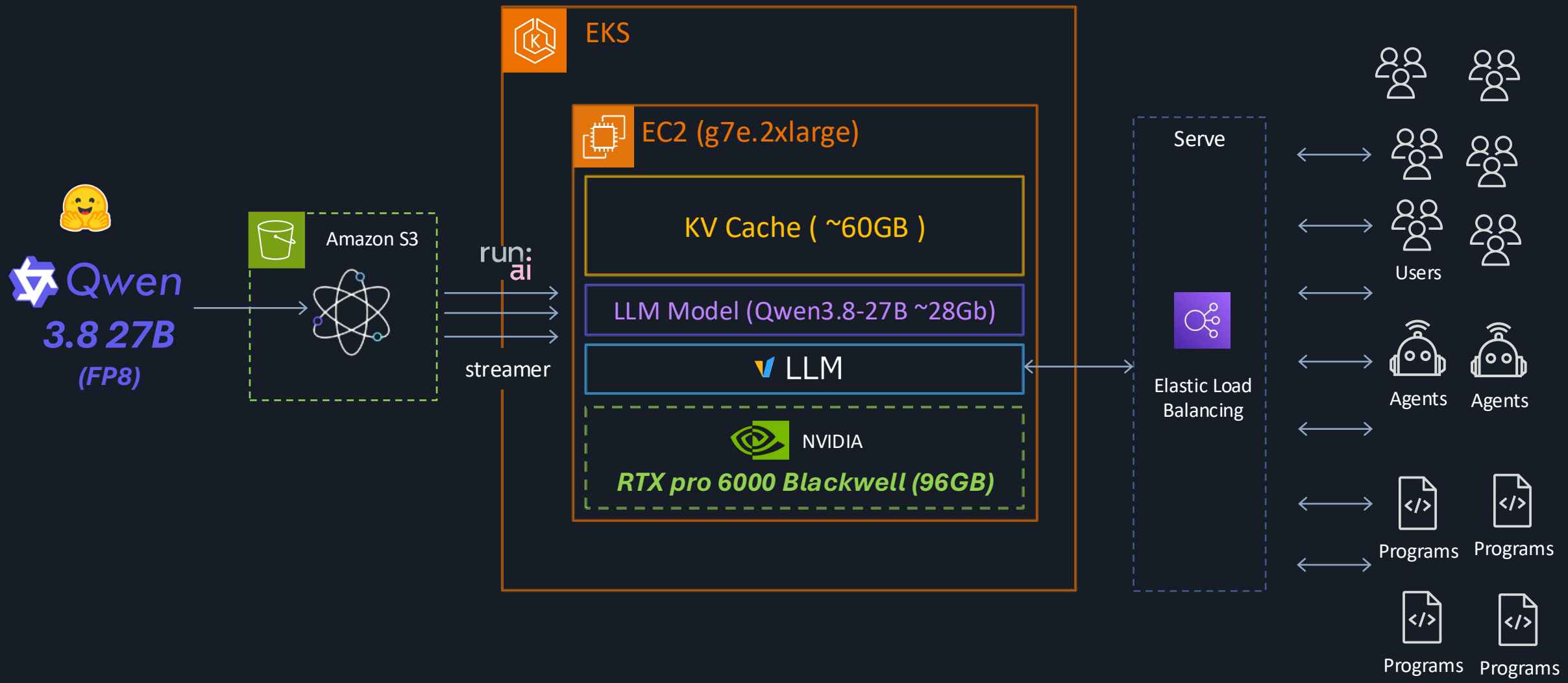
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Demo

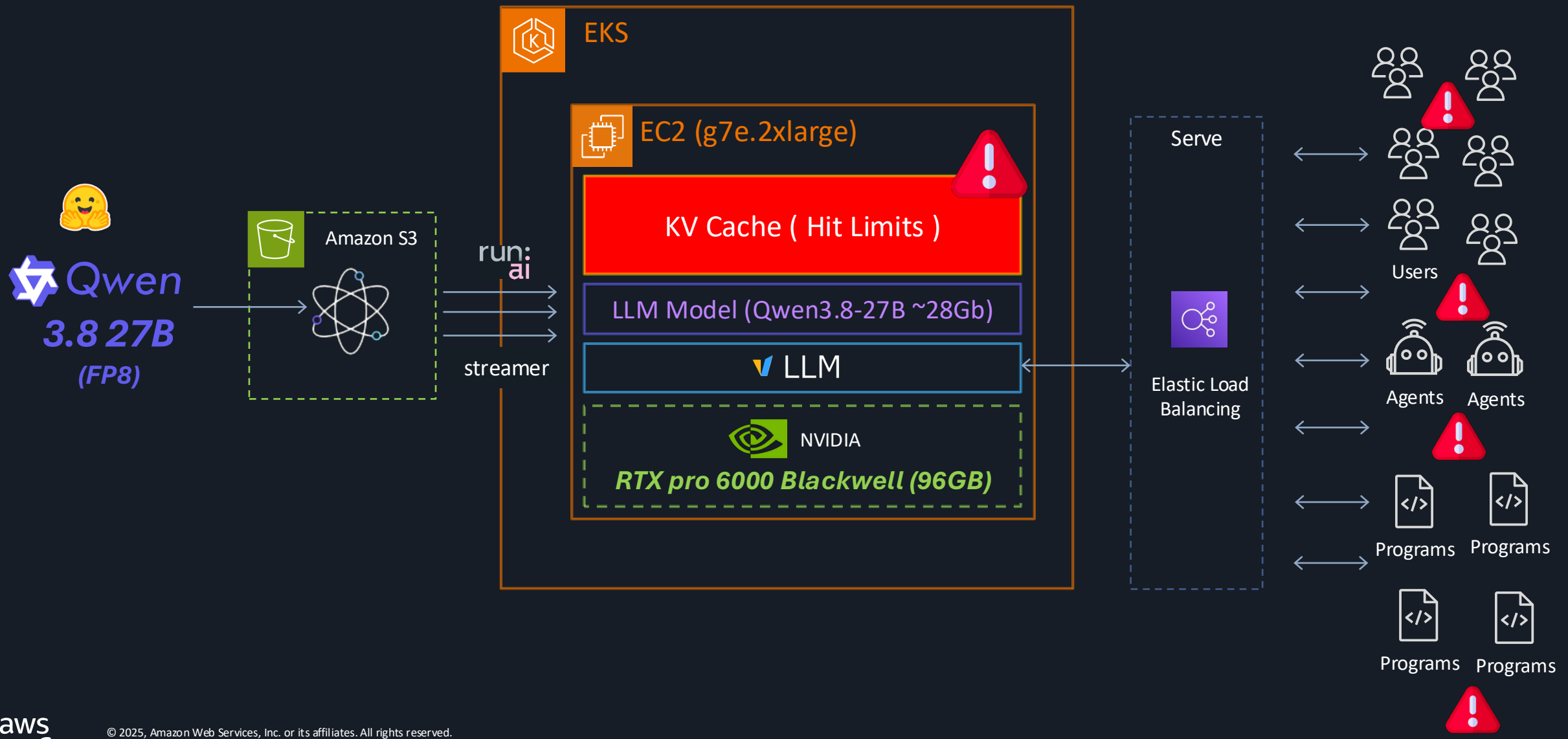
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Single Node



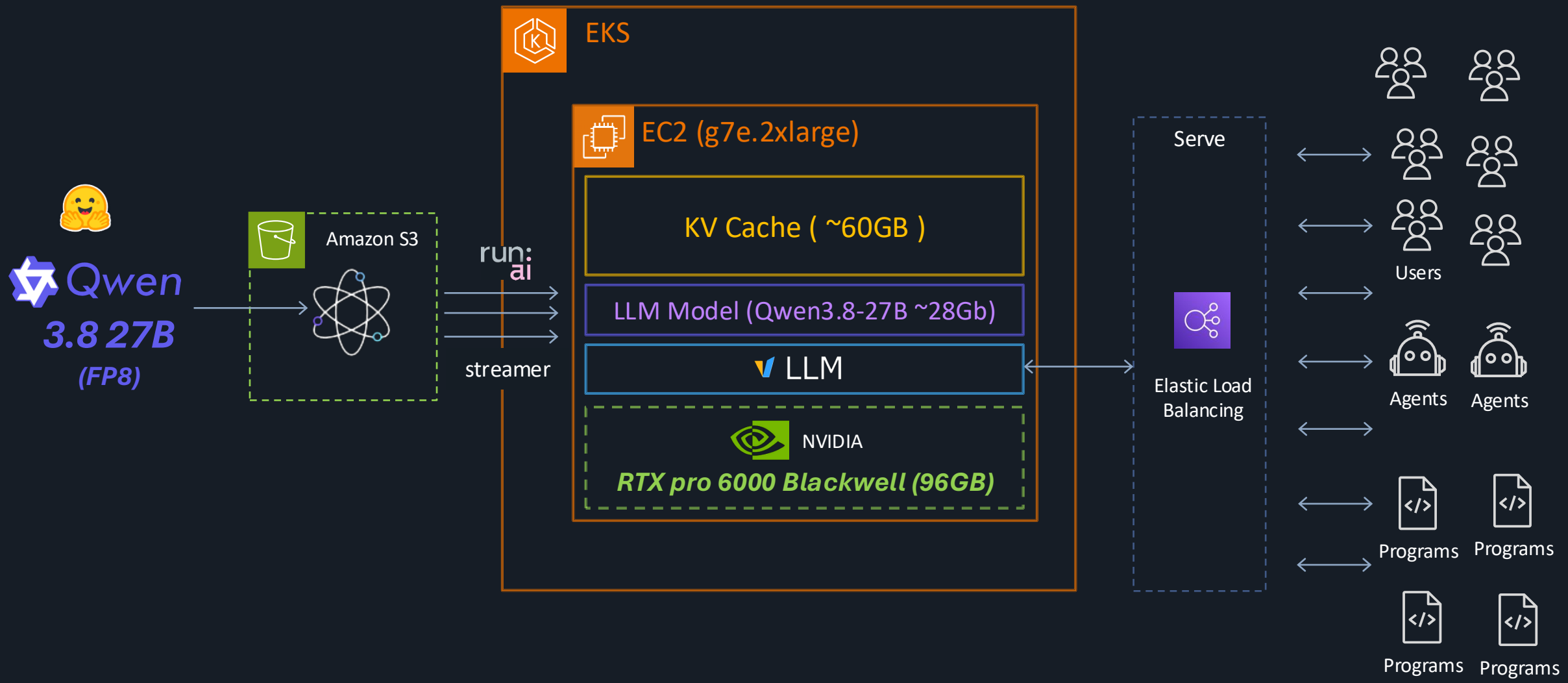
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Single Node

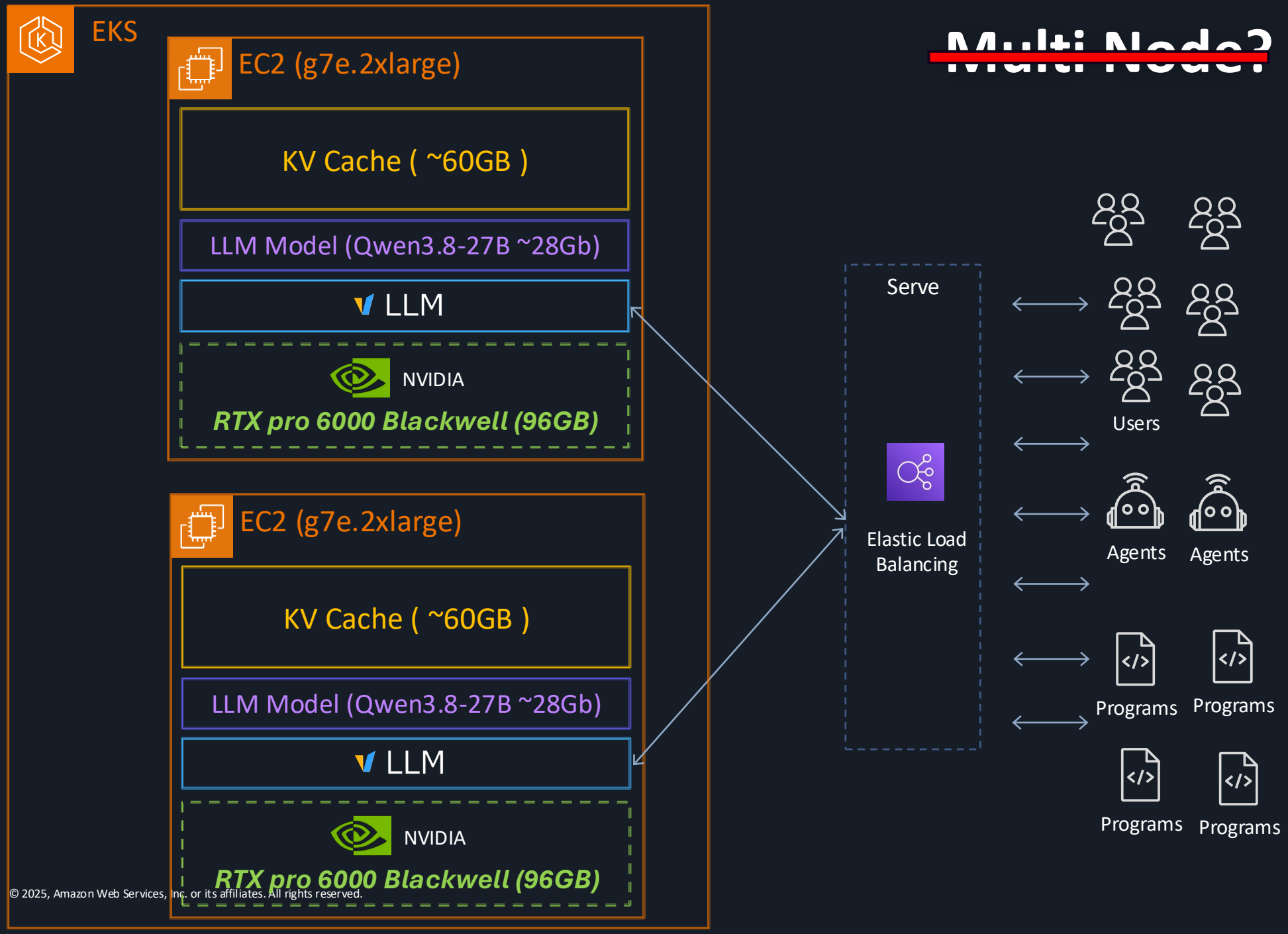


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Single Node



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~~Multi Node?~~

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Understand AI Inference Process

POST /v1/chat/completions
-d “message” [{…}]



Prefill

Process prompt (initial KV cache)

(TFLOPS – TTFT)

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Understand AI Inference Process

Prefill

Process prompt (initial KV cache)

(TFLOPS – TTFT)



Decode

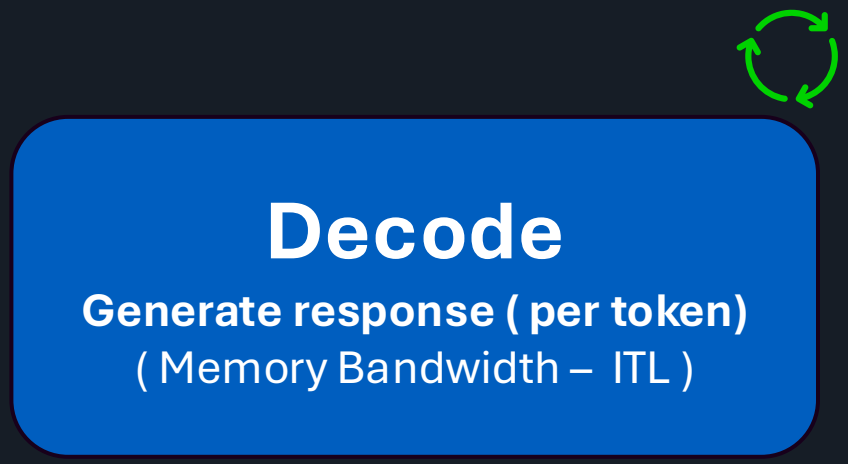
Generate response (per token)

(Memory Bandwidth – ITL)

aws

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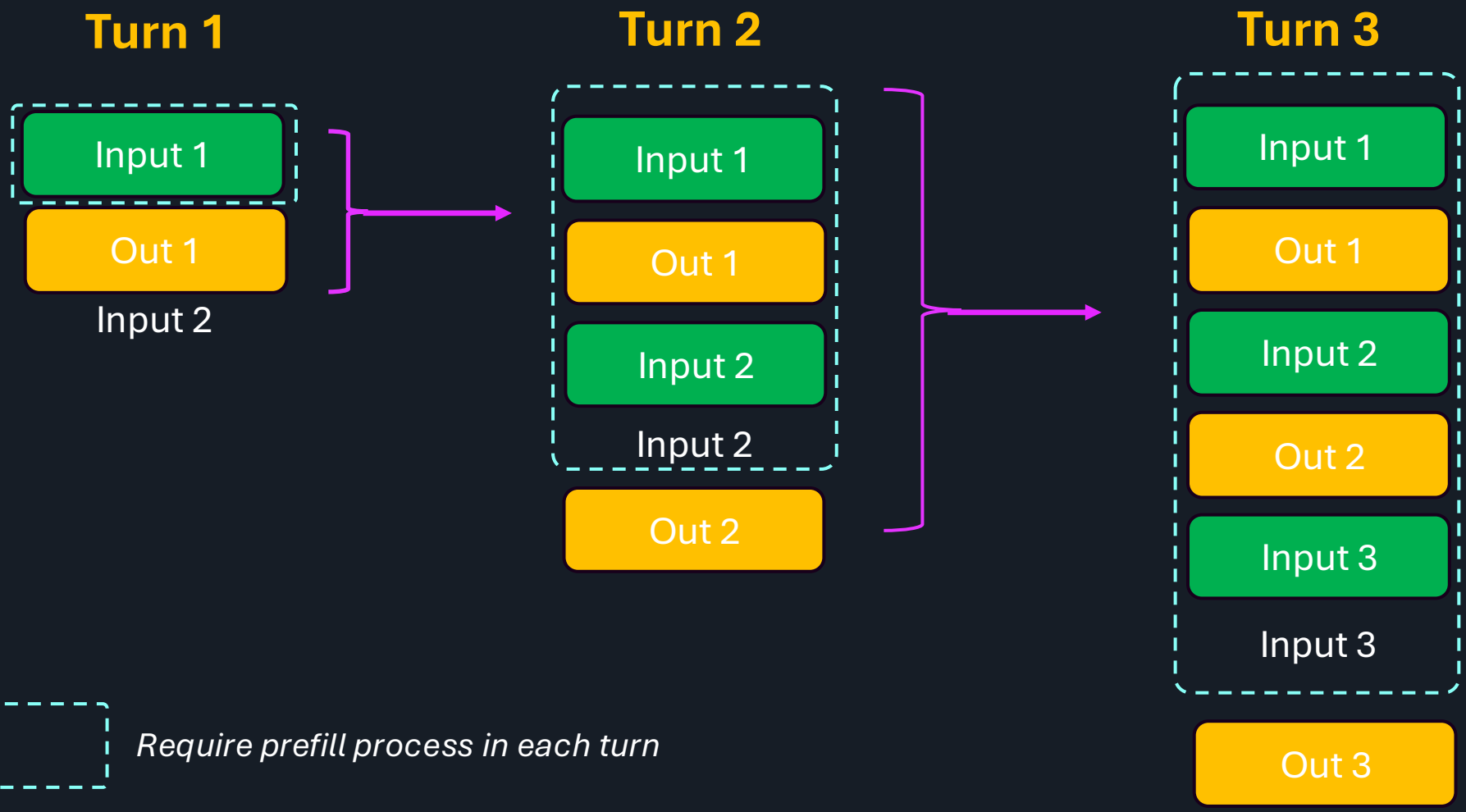
Understand AI Inference Process



Cloud Native techn ologies empower organ izations to scale . <|endoftext|>

aws
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Multi-turn Conversation work



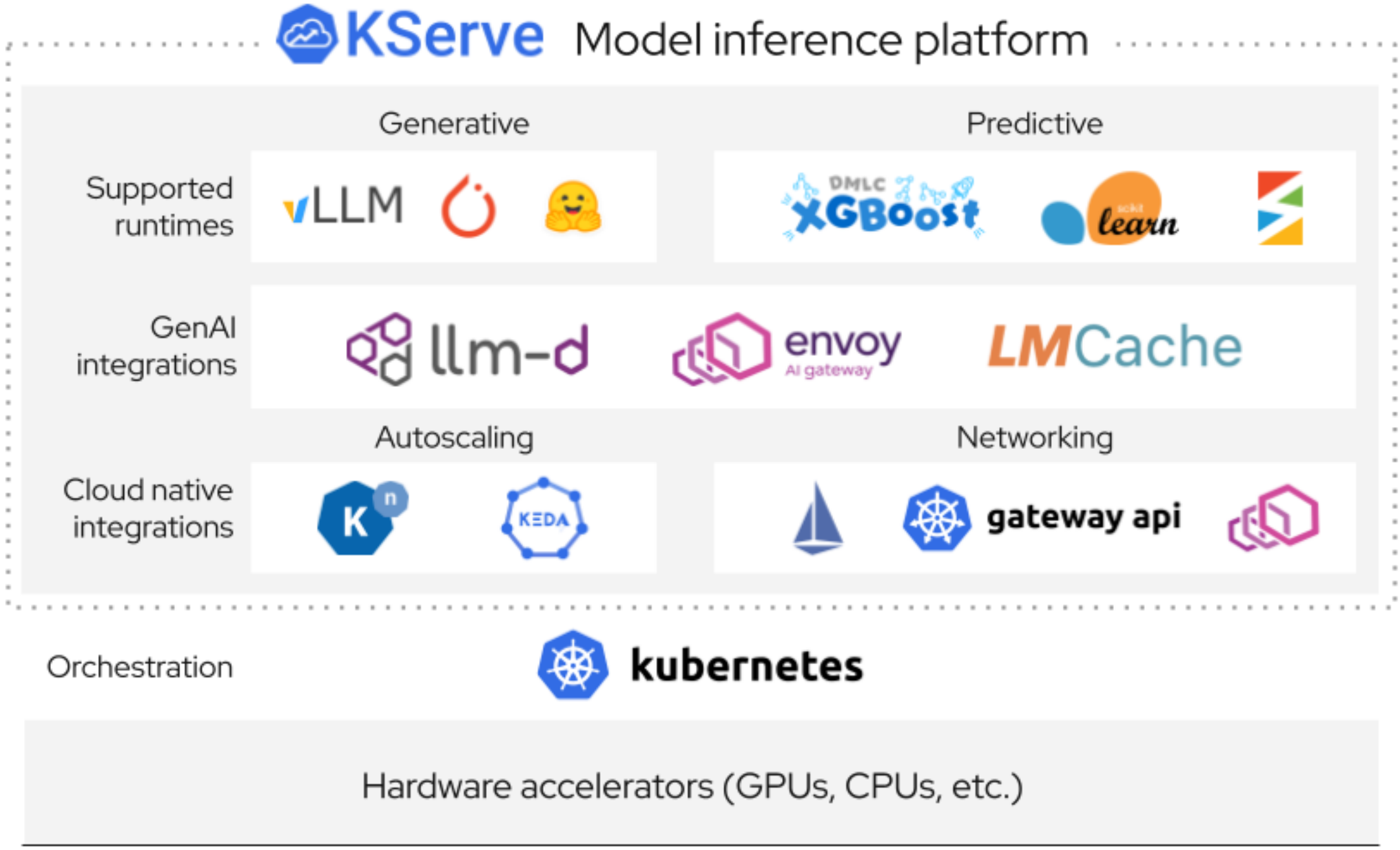
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Standardized Distributed Generative and Predictive AI Inference Platform for Scalable, Multi-Framework Deployment on Kubernetes

aws

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KServe
Model inference platform
Generative
Predictive
Supported runtimes
vLLM
DMLC XGBoost
scikit learn
GenAI integrations
llm-d
envoy AI gateway
LMCache
Autoscaling
Networking
Cloud native integrations
KEDA
gateway api
Orchestration
kubernetes
Hardware accelerators (GPUs, CPUs, etc.)

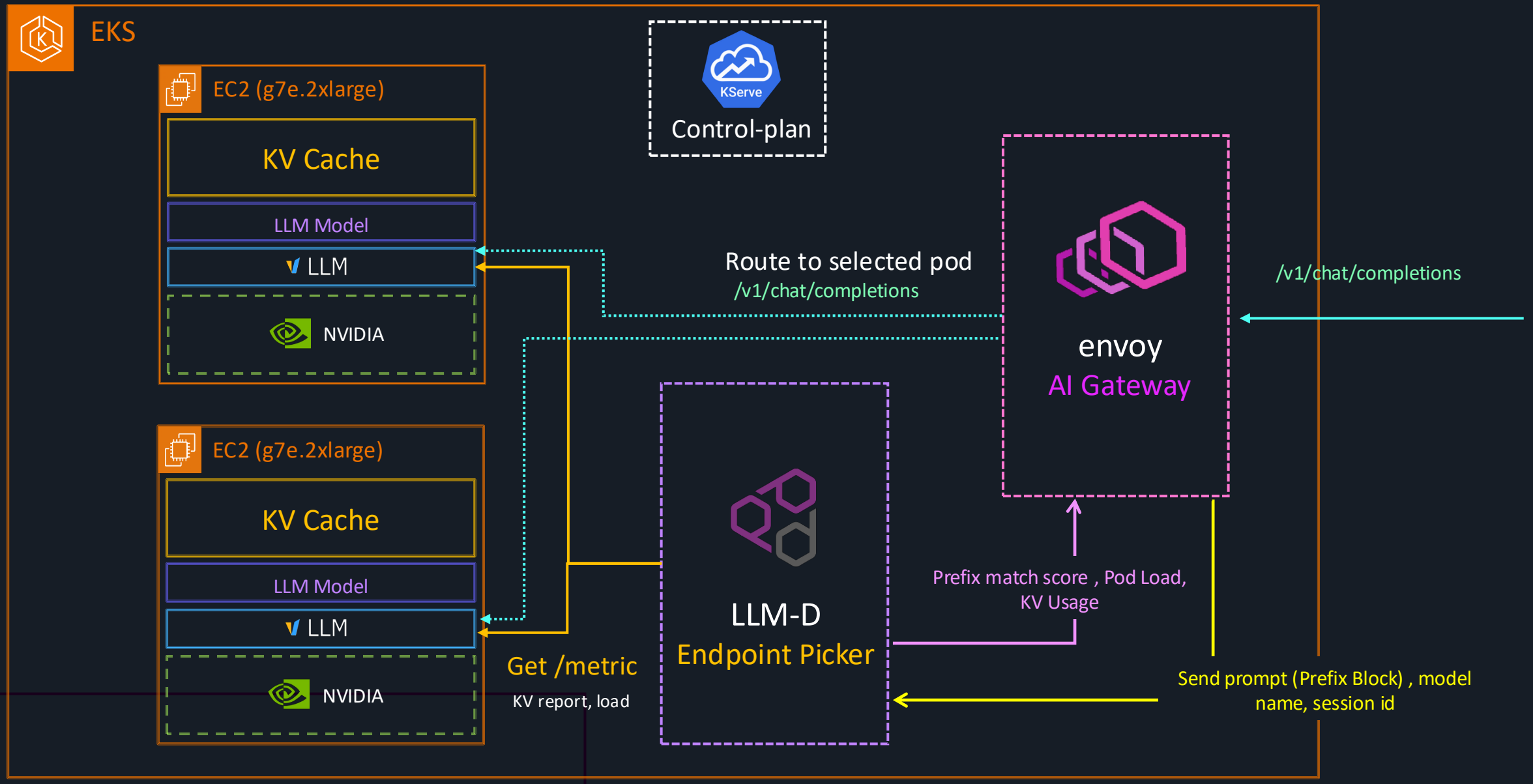
vLLM Standalone vs. KServe Managed

- **Autoscale** : Serverless & Traffic-Driven Autoscaling (KEDA/Knative)
- **Prefix-cache-aware routing** : Context-Aware / Prefix-Cache Aware Routing
- **Prefill/decode disaggregation :** Distributed Prefill/Decode Disaggregation
- **Multi-node (LeaderWorkerSet) :** Multi-Node Scale-Out via LeaderWorkerSet (Tensor Parallelism / Pipeline Parallelism)
- **KV Cache Offloading (LMCache)** : Distributed KV Cache Management & Offloading (LMCache)

aws

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Demo : Prefix-cache-aware routing



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